

$$\mathcal{L}_{SM} = \mathcal{L}_{gauge} + \mathcal{L}_{fermions} + \mathcal{L}_{Yukawa} + \mathcal{L}_{Higgs}$$

$$\begin{aligned}\mathcal{L}_{gauge} &= \mathcal{L}_{gluon} - \mathcal{L}_{weak} - \mathcal{L}_{supercharge} \\ &= -\frac{1}{4}(\partial_\mu G_\nu^a - \partial_\nu G_\mu^a)(\partial^\mu G^{a\nu} - \partial^\nu G^{a\mu}) \\ &\quad - \frac{g_s}{2}f^{abc}(\partial_\mu G_\nu^a - \partial_\nu G_\mu^a)G^{b\mu}G^{c\nu} \\ &\quad - \frac{g_s^2}{4}f^{abc}f^{ade}G_\mu^b G_\nu^c G^{d\mu}G^{e\nu} \\ &\quad - \frac{1}{4}(\partial_\mu W_\nu^i - \partial_\nu W_\mu^i)(\partial^\mu W^{i\nu} - \partial^\nu W^{i\mu}) \\ &\quad - \frac{g}{2}\epsilon^{ijk}(\partial_\mu W_\nu^i - \partial_\nu W_\mu^i)W^{j\mu}W^{k\nu} \\ &\quad - \frac{g^2}{4}\epsilon^{ijk}\epsilon^{ilm}W_\mu^j W_\nu^k W^{l\mu}W^{m\nu} \\ &\quad - \frac{1}{4}(\partial_\mu B_\nu - \partial_\nu B_\mu)(\partial^\mu B^\nu - \partial^\nu B^\mu)\end{aligned}$$

$$\begin{aligned}\mathcal{L}_{SM} &= -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} - \frac{1}{4}W_{\mu\nu}^i W^{i\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} \\ &+ \bar{\psi}_L i\gamma^\mu D_\mu \psi_L + \bar{\psi}_R i\gamma^\mu D_\mu \psi_R \\ &+ -(y_{ij}\bar{\psi}_{L,i}\Phi\psi_{R,j} + \text{h.c.}) \\ &+ (D_\mu\Phi)^\dagger(D^\mu\Phi) - \mu^2\Phi^\dagger\Phi - \lambda(\Phi^\dagger\Phi)^2 \\ &+ \text{gauge-fixing terms} + \text{ghost terms} + \theta\text{-terms} + \text{neutrino mass terms (if included)}\end{aligned}$$